



Unix Engine: Technical User Manual

Version: 1.2

Host Requirement: Docker Desktop for Windows

1. Prerequisites (Must-Have)

Before attempting to use the engine, the user **must** have the following installed and running:

- **Docker Desktop:** The engine runs as a containerized service. Docker Desktop must be active (using either the WSL2 or Hyper-V backend).
 - **PowerShell Core:** Required for the `deploy.ps1` automation script.
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2. Installation & Setup

1. **Launch Docker Desktop:** Ensure the Docker icon is visible in your system tray and the service is "Running."

Build the Factory: Open a terminal in the project root and run:

```
PowerShell  
docker build -t unix-compiler .
```

2.

Start the Engine:

```
PowerShell  
docker run -d -p 8050:80 --name brave_snyder unix-compiler
```

3.

3. The Workspace (Where the App Lives)

The internal folder **/workspace** is the heart of the engine.

- **Build Site:** This is where your code is unzipped and compiled.
- **Final Home:** After a successful build, the engine wipes all source files (**.c**, **.h**, **Makefile**) to keep the environment clean.
- **Finding the Binary:** You will find your final **{ProjectName}.out** file sitting alone in the **/workspace** directory.

Manual Access: To run or inspect the app inside the container:

PowerShell

```
docker exec -it brave_snyder bash
```

```
cd /workspace
```

```
./YourProjectName.out
```

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4. Usage: The Remote Control

Move **deploy.ps1** to your project folder and execute it:

PowerShell

```
./deploy.ps1
```

The script automatically:

1. Detects your folder name for the final binary name.
2. Packages only essential source files (excluding **.vscode**, **.git**, etc.).
3. Ships the code to the engine and retrieves the compiled Linux binary.

5. Troubleshooting

- **"Status 000":** Docker Desktop is likely not running, or the container **brave_snyder** has stopped.
- **"413 Request Entity Too Large":** You are likely trying to send the **.vscode** folder. Ensure the exclusions in the script are active.